



A witness for every balance break.

A sensor station for the Flamingo balance test, with the teacher in charge.



Sensors flag

Beam + held-foot insole



Teacher decides

Confirm, dismiss or add



Record keeps both

Timestamp + decision

Insole + app built

Beam + hub planned



Balance beam, sensor insole and teacher app

Problem Statement ID - 26213

Problem Statement Title-

Student Innovation-Ideas that can boost fitness activities and assist in keeping fit.

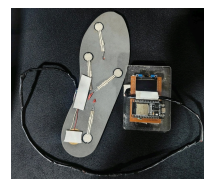
Theme- Fitness & Sports PS Category- Hardware

Team ID- 121295

Team Name (Registered on portal)-

Team Palanteen

Thakur Shyamnarayan Engineering College



Our MVP prototype
Insole sensing core



NIVA SAKSHI

Proposed Solution (Describe your Idea/Solution/Prototype)

Every balance break gets a timestamp and a human decision.

02 / THE EXPERIENCE

Detailed explanation of the proposed solution

TODAY · THREE JOBS AT ONCE



Watch. Pause. Write the count.
Manual timing and fall tally.



1 · Balance

One leg on the beam.
Clock runs.



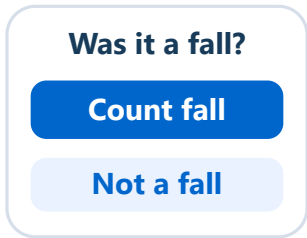
2 · Flag

Touchdown or step-off
creates an observation.



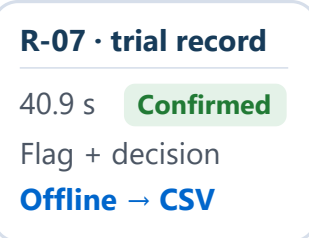
3 · Pause

Clock pauses at the
event timestamp.



4 · Decide

Teacher confirms,
dismisses or adds a fall.

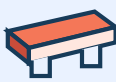


5 · Keep

Save every flag and
the teacher's decision.

Zero-check first · resume when balance returns · continue to 60 s of balancing

How it addresses the problem



Standing foot steps off

Beam unloading flags it · planned



Held foot touches down

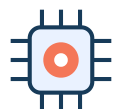
Insole contact flags it · core built



Foot released without touchdown

Teacher sees it and adds the fall

Innovation and uniqueness of the solution



Two sensing witnesses



Human confirmation



Flag + decision history



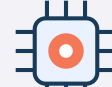
No child video

TECHNICAL APPROACH

Technologies to be used (e.g. programming languages, frameworks, hardware)

Two contact signals feed one teacher-led record.

03 / INSIDE THE STATION

 **ESP32 · C++**
Contact edges + timestamps

 **Zero-check gate**
Invalid sensing → invalid trial

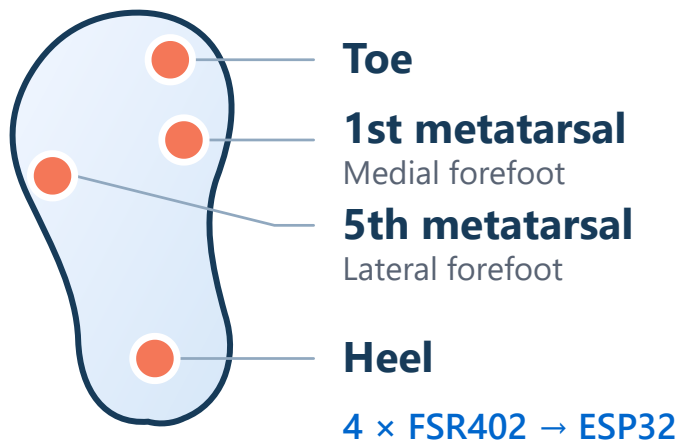
 **Flutter / Dart**
Protocol engine + teacher call

 **Local storage + CSV**
Flag source + decision saved

Methodology and process for implementation (Flow Charts/Images/ working prototype)

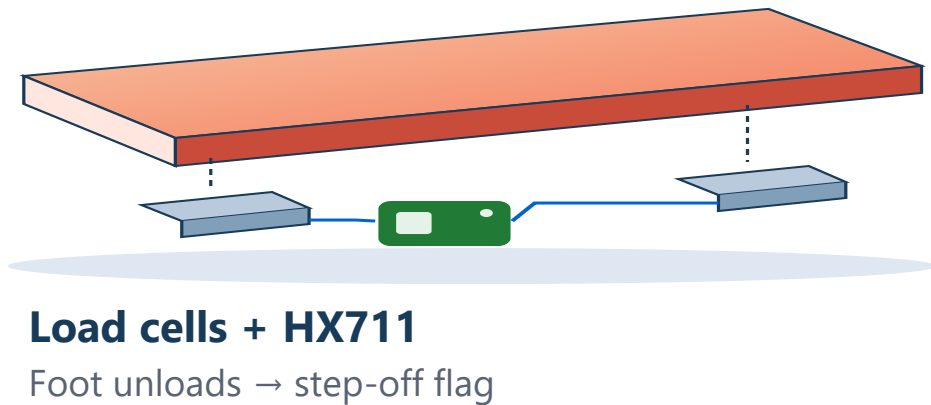
Insole · held foot

Core built



Beam · standing foot

Planned



Insole · built

current direct route

Teacher phone

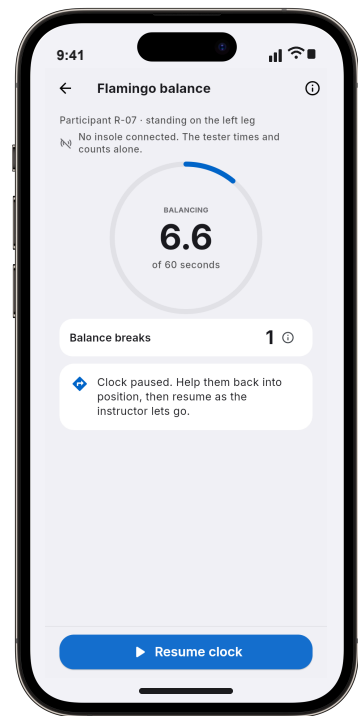
Pause → review → resume

Trial record

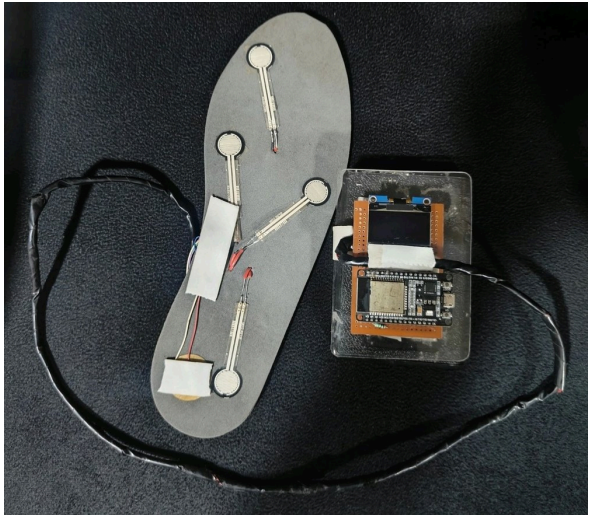
Flag + decision → CSV

Beam + hub · planned

Our app + MVP prototype



Pause + resume



Our MVP prototype

Insole → phone
Android APK builds

Next: full station trial

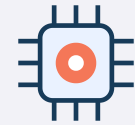
Insole: 4 × FSR402 + MPU6050 + heel piezo. Beam/hub: load cells + HX711 + shared clock, proposed. FSRs measure contact timing, not body-weight force.

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

Build the station. Prove the timing. Then take it to schools.

04 / THE PATH TO PROOF



1 · Core
Insole + app + APK
Built



2 · Station
Beam + hub integration
To build



3 · Prove
Adult trial + tester study
To measure



4 · School
Usability + consented pilot
Proposed

Potential challenges and risks

Strategies for overcoming these challenges



Beam and complete live station unfinished

Standard parts + filmed adult trial.
Build beam/hub and test on the intended phone.



Missed fall, drift or lost zero

Teacher adds missed events; zero-check gates scoring.
A sensing fault marks the trial invalid.



Accuracy and time per child unknown

Compare with a trained tester + slow-motion video.
Measure agreement, setup time and battery life.

 **Safety by design:** teacher/spotter present · no child video · IDs only · school consent before a pilot

 **One shared kit per school**
Economics to validate

 +  + 

Insole Beam Hub

Include power, enclosure, assembly and service before quoting a full-kit price.



Insole prototype at Automation Expo, Mumbai.

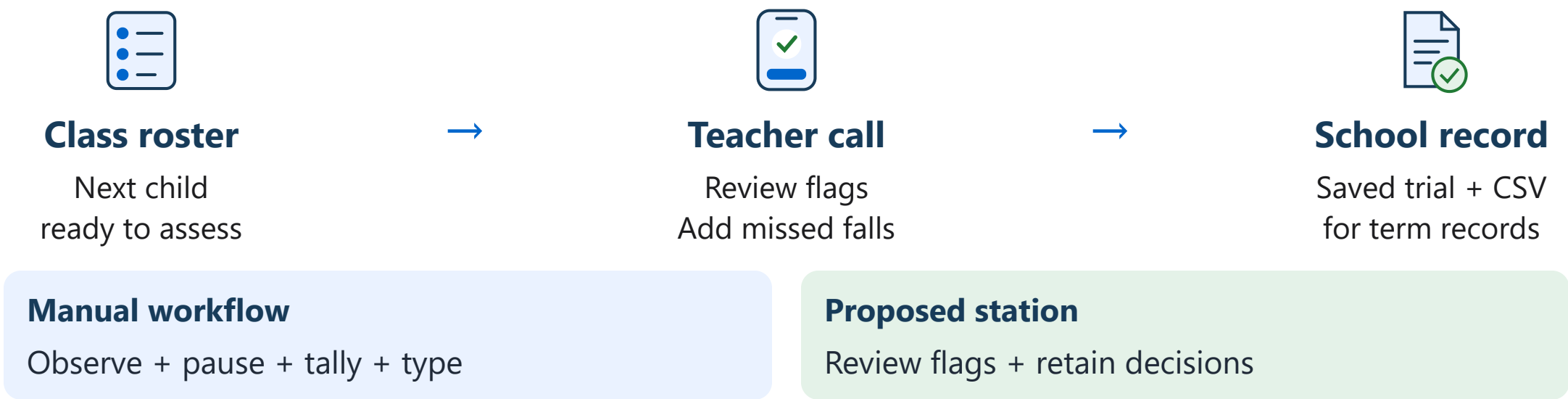
IMPACT AND BENEFITS

Potential impact on the target audience

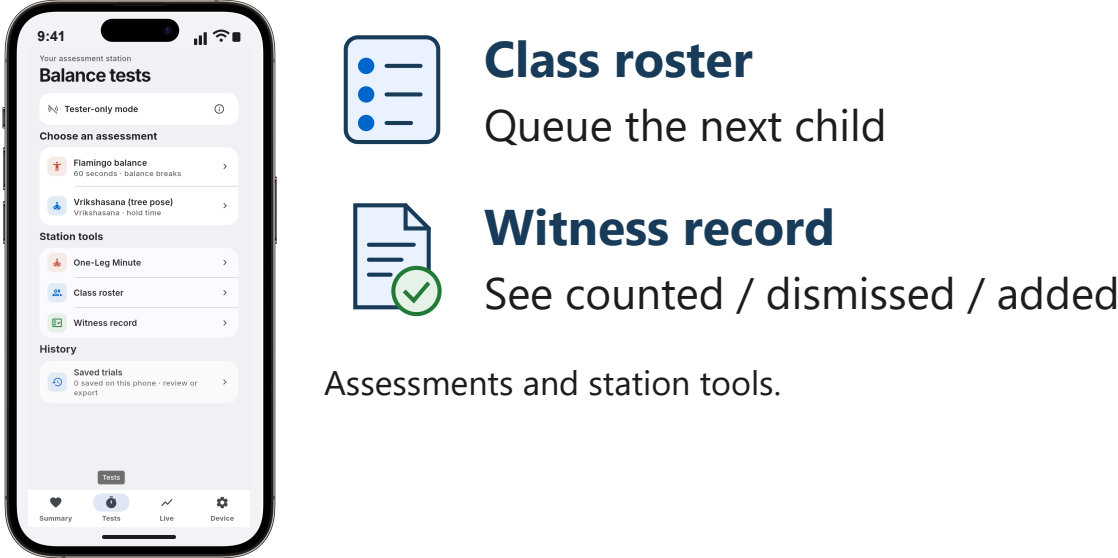
From one child’s balance test to the whole class.

05 / SCHOOL DAY, SIMPLIFIED

A record that follows the assessment



The app already speaks this language



4.5 lakh+ schools awarded Fit India School Flags [3]
PIB · June 2026

 **One-Leg Minute:** practise in PE → track own best → keep moving.
Practice in PE · track personal progress.

Benefits of the solution (social, economic, environmental, etc.)

-  **Social · no video**
Teacher retains authority
-  **Economic · shared kit**
Cost per child to measure
-  **Environmental**
Rechargeable; serviceable
-  **Strategic · traceable**
Record shows each decision

RESEARCH AND REFERENCES

Details / Links of the reference and research work

Research that shapes the station.

06 / RESEARCH → DESIGN



Timing protocol [1]

Pause on balance loss.

→ **Event flag** → **clock pauses**



Loss definitions [1]

Step-off and held-foot release.

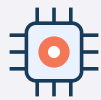
→ **Beam + insole + teacher**



School assessment [1, 2]

Teachers assess; schools retain results.

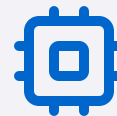
→ **Human call + term CSV**



FSR402 datasheet [11]

Force range is limited.

→ **Contact timing, not force**



HX711 datasheet [10]

Load-cell ADC supports the beam.

→ **Step-off detector · planned**



Validation plan [9, 15]

Compare with a trained tester.

→ **Measure agreement + time saved**

SOURCE LIBRARY · CLICKABLE REFERENCES

1. Fit India Mission, *Fitness Protocols and Guidelines* (5-18 and 18-65 years), Ministry of Youth Affairs & Sports with Ministry of Health & Family Welfare. fitindia.gov.in/fitnessprotocols
2. Fit India School certification criteria. fitindia.gov.in/fit-india-school
3. PIB backgrounder, 13 June 2026: more than 4.5 lakh Fit India School Flags. [static.pib.gov.in \(PDF\)](https://static.pib.gov.in/PIB/mediarelease/2026/20260613/fit-india-school-flags.pdf)
4. Sports Authority of India press release, 31 Dec 2019, Khelo India app assessments. [sportsauthorityofindia.nic.in \(PDF\)](https://sportsauthorityofindia.nic.in/press-releases/2019/12/31/khelo-india-app-assessments.pdf)
5. CBSE, Physical Education curriculum (Class XI-XII), 2026-27. [cbseacademic.nic.in \(PDF\)](https://cbseacademic.nic.in/Physical-Education-Curriculum-Class-XI-XII-2026-27.pdf)
6. Khelo India (School Version) assessor app listing. [play.google.com](https://play.google.com/store/apps/details?id=com.kheloindia.assessment), kheloindia.com/assessment
7. SIH 2026 Guidelines, incl. evaluation criteria and the originality clause. [sih.gov.in/letters/2026 \(PDF\)](https://sih.gov.in/letters/2026/SIH-Guidelines-2026.pdf)
8. SIH 2026 Idea Presentation Format. [sih.gov.in/letters/2026 \(PPTX\)](https://sih.gov.in/letters/2026/SIH-Idea-Presentation-Format.pptx)
9. LBRCE, SIH 2024 final report. [lbrce.ac.in \(PDF\)](https://lbrce.ac.in/SIH-2024-Final-Report.pdf)
10. Avia Semiconductor, HX711 24-bit ADC for weigh scales, datasheet. [cdn.sparkfun.com \(PDF\)](https://cdn.sparkfun.com/datasheets/Avia/HX711-24-bit-ADC-for-weigh-scales.pdf)
11. Interlink Electronics, FSR 402 datasheet. [cdn.sparkfun.com \(PDF\)](https://cdn.sparkfun.com/datasheets/Interlink/FSR-402-datasheet.pdf)
12. Probots, HX711 amplifier board. [probots.co.in](https://probots.co.in/hx711-amplifier-board)
13. Probots, 80 kg bar load cell. [probots.co.in](https://probots.co.in/80-kg-bar-load-cell)
14. BTrackS Assess Balance System, list price (a force plate, cited for price only). balancetrackingsystems.com
15. Moticon OpenGo price list (EUR), prior art. [moticon.com \(PDF\)](https://moticon.com/OpenGo-price-list.pdf). KINVENT K-Force price: kinvent.com product page.